

DS 4XX

**Understanding Economy, Society and Policy: A Data-Driven Approach
Autumn 2025**

Instructor: Dr. Parul Gupta

Commented [PG1]: Course code to be confirmed

Course Placement: DS 4XX is a technical elective/open elective for Second Year (Semester 3) MSc Data Science/MSc IT students.

Course Format: 3 hours of lectures and 2 hours of lab each week (3-0-2-4)

Pre-requisites: None

Course Content: This course introduces students to foundational economic concepts in macroeconomics and development economics. The interconnections and complementarities between economics, society and government policies will be discussed. Students will be encouraged to recognise and appreciate the two-way relationship between economics and policy-making. The course will focus on economic reasoning through the analysis and interpretation of real-world data. Lectures will be supplemented with guided empirical analysis of real-world economic problems.

Course Material (Selected):

1. *Doing Economics*, CORE Team
E-book available online: <https://www.core-econ.org/doing-economics/book/text/0-3-contents.html>
2. *Economy, Society and Public Policy*, CORE Team
E-book available online: <https://www.core-econ.org/espp/book/text/0-3-contents.html>
3. *Basic Economics: A Citizen's Guide to the Economy*, Thomas Sowell (5th edition). ISBN: 9780465081455

Data-based exercises will be completed using MS Excel/Google Sheets with the help of publicly available datasets.

Method of Assessment:

Data exercises (30 %)

Mid-term exam (30 %)

Group project and presentation (40 %)

Plagiarism on any assessment component will result in a failing grade for the course.

Course Outcomes: On successfully completing the course, students will have a basic familiarity with economic reasoning and global socio-economic concerns. They will become aware of publicly accessible sources to obtain real-world macroeconomic data. They will be able to analyse and interpret data and communicate their findings in written and verbal formats. In doing so, they will be able to enhance their writing, presentation and analytical skills.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
	X		X	X	X	X		X	X		

Lecture Schedule

S. No.	Description	Number of lectures
1	Understanding what economics is (and what it is not) Introducing basic concepts in economics such as opportunity costs, incentives and trade-offs	2
	Discussing the connections between economics, society and policies	
2	Understanding prosperity and well-being Defining GDP, GDP/capita, economic growth, HDI. Critiquing these measures.	6
	Lab exercise: Identifying trends in GDP, economic growth and HDI for various countries	
3	Understanding poverty and inequality Measuring poverty, inequality using alternative methods. Interpreting Lorenz curves and GINI coefficients	4
	Lab exercise: Plotting Lorenz curves and interpreting them for various countries	
4	Understanding unemployment and labour market frictions Discussing the different types of unemployment and their implications for society and the individual	4
	Discussing policies to address unemployment	
	Lab exercise: Discovering the relation between work status and life satisfaction	
5	Understanding global socio-economic concerns Discussing the causes and implications of global concerns such as environmental degradation and conflict	6
	Lab exercise: Investigating trends in global temperatures and carbon emissions	
6	Understanding the motivation and implications of government policies Introducing taxes, subsidies and transfer payments.	5
	Discussing the unintended consequences of these policies.	
	Lab exercise: Critiquing the cash handout program of the Hong Kong government	
7	Understanding the banking and credit sector Introducing the role of banks and the flow of credit in the economy.	6
	Recognising the constraints faced by households in accessing credit	
	Lab exercise: Comparing banking systems across the world	
	Lab exercise: Analysing the determinants of credit access for households	